数 学 作 业 纸

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1.6. $dF = q\,ds = qr\,d\theta$.

$dF_x = dF\sin\theta = qr\sin\theta\,d\theta$

$dF_y = -dF\cos\theta = -qr\cos\theta\,d\theta$

$F_x = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} qr\sin\theta\,d\theta = qr[-\cos\theta]_{-\frac{\pi}{6}}^{\frac{\pi}{6}} = 0$.

$F_y = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} -qr\cos\theta\,d\theta = -qr[\sin\theta]_{-\frac{\pi}{6}}^{\frac{\pi}{6}} = -qr\left(\sin\frac{\pi}{6} - \sin\left(-\frac{\pi}{6}\right)\right) = -2qr\sin\left(\frac{\pi}{6}\right)$

$F_y = -2\times8\times50\cdot\sin30^\circ = -400N$.

$M_O = \int r\,dF = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} r(qr\,d\theta) = qr^2\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} d\theta = qr^2\left(\frac{\pi}{3}\right)$.

$M_O = 8\times50^2\times\frac{\pi}{3} = \frac{20000\pi}{3}\ N\cdot m$

$F_R\cdot x_C = M_O$　$x_C = \frac{50\pi}{3} = 52.5$.

合力 400N 竖直向下，$d = 52.5\ cm$.

1.12 (d). $F_{Ax} = 0$

$F_{Ay} - ql - F = 0$.

$F_{Ay} = F + ql$.

$M_A - Fl - ql\frac{l}{2} = 0$.

$M_A = Fl + \frac{1}{2}ql^2$ (逆时针).

1.19. 对D. $\Sigma F_x = F - F_{N3} = 0$.

$F_{N3} = F$

$\Sigma F_y = -F_{N2} = 0$.

$F_{N2} = 0$.

对L. $\Sigma F_x = F_{N3} + F_{N5}\cos45^\circ = 0$

代入 $F_{N3} = F$

$F + F_{N5}\frac{\sqrt{2}}{2} = 0$.

$F_{N5} = -\sqrt{2}F$.

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$\sum F_y = -F_{N4} - F_{N5}\sin 45^\circ = 0.$

$-F_{N4} - (-\sqrt{2}F)\frac{\sqrt{2}}{2} = 0.$

$F_{N4} = F.$

对B. $\sum F_x = -F_{N1} - F_{N5}\cos 45^\circ = 0.$

$-F_{N1} - (-\sqrt{2}F)\frac{\sqrt{2}}{2} = 0$

$F_{N1} = F$

综上. $F_{N1} = F$ (拉)

$F_{N2} = 0$

$F_{N3} = F$ (拉)

$F_{N4} = F$ (拉)

$F_{N5} = -\sqrt{2}F$ (压).

1.12(b)

$\sum F_x = 0 \Rightarrow F_{Ax} = 0.$

$\sum M_A = M_e - Fa + F_{By}(2a) - qa \cdot 2.5a = 0.$

$qa^2 - qa \cdot a + 2aF_{By} - 2.5qa^2.$

$F_{By} = 1.25qa.$ 向上.

$\sum F_y = F_{Ay} - F + F_{By} - qa = 0.$

$F_{Ay} - qa + 1.25qa - qa = 0.$

$F_{Ay} = 0.75qa$ 向上.

(f) $\sum F_x = 0 \Rightarrow F_{Ax} = 0.$

$\sum M_A = M_e - qa \cdot 1.5a + F_{By} \cdot 2a = 0.$

$qa^2 - 1.5qa^2 + 2aF_{By} = 0$

$F_{By} = 0.25qa$ 向上.

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$\Sigma F_y = F_{Ay} - qa + F_{By} = 0$.

$F_{Ay} - qa + 0.25qa = 0$.

$F_{Ay} = 0.75qa$ 向上.

1.14 (a) $F_{Ax} + F = 0$ $F_{Ax} = -F$. 向左.

$-F(2a) - F(1.5a) + F_{By}(3a) = 0$

$-3.5Fa + 3a F_{By} = 0$

$F_{By} = \frac{7}{6}F$ 向上.

$F_{Ay} - F + F_{By} = 0$.

$F_{Ay} - F + \frac{7}{6}F = 0$

$F_{Ay} = -\frac{1}{6}F$ 向下.

1.26. $M_P = Pr = 10kN \times 100mm = 1000 kNmm$

$M = F_1 R - F_2 R = (F_1 - F_2)R$.

$(F_1 - F_2)R = Pr$.

$(2F_2 - F_2) \cdot 200 = 1000$

$F_2 = 5kN$.

$F_1 = 10kN$.

$F_{1x} = F_1 \cos 30 = 5\sqrt{3} kN$.

$F_{1y} = -F_1 \sin 30° = -5kN$.

$F_{2x} = F_2 \cos 30° = 2.5\sqrt{3} kN$

$F_{2y} = F_2 \sin 30° = 5 \times 0.5 = 2.5 kN$.

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$F_y = F_{1y} + F_{2y} = -5 + 2.5 = -2.5kN$.　$P = 10kN$.

$-P \cdot 300 - |F_y| \cdot 600 + F_{By} \cdot 1000 = 0$.

$-3000 - 1500 + 1000F_{By} = 0$　$F_{By} = 4.5k$ 向上.

$F_{Ay} + F_{By} - P - |F_y| = 0$　$F_{Ay} + 4.5 - 10 - 2.5 = 0$

$F_{Ay} = 8kN$ 向上.

$F_z = F_{1z} + F_{2z} = 5\sqrt{3} + 2.5\sqrt{3} = 7.5\sqrt{3}kN$ 向左.

$F_z \cdot 600 + F_{Bz} \cdot 1000 = 0$.

$7.5\sqrt{3} \cdot 600 + 1000F_{Bz} = 0$.

$F_{Bz} = -4.5\sqrt{3}kN$ 向右.

$F_{Az} + F_{Bz} + F_z = 0$

$F_{Az} = -3\sqrt{3}\ kN$ 向右.